

6.1.2 Carbonyl compounds

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
Reactions of carbonyl compounds	
(a) oxidation of aldehydes using $\text{Cr}_2\text{O}_7^{2-}/\text{H}^+$ (i.e. $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}_2\text{SO}_4$) to form carboxylic acids	In equations for organic redox reactions, [O] and [H] should be used. PAG7 (see also 6.3.1 c)

- (b) nucleophilic addition reactions of carbonyl compounds with:
- (i) NaBH_4 to form alcohols
 - (ii) HCN [i.e. $\text{NaCN(aq)}/\text{H}^+(\text{aq})$], to form hydroxynitriles (**see also 6.2.4 b**)
- (c) the mechanism for nucleophilic addition reactions of aldehydes and ketones with NaBH_4 and HCN

For NaBH_4 , the nucleophile can be considered as being the hydride ion, H^- , with subsequent protonation of the organic intermediate from H_2O .

For HCN , initial nucleophilic attack is from CN^- ions; subsequent protonation stage can be shown using H_2O or H^+ .

HSW1,2,8 Use of reaction mechanisms to explain organic reactions.

Characteristic tests for carbonyl compounds

- (d) use of 2,4-dinitrophenylhydrazine to:
- (i) detect the presence of a carbonyl group in an organic compound
 - (ii) identify a carbonyl compound from the melting point of the derivative
- (e) use of Tollens' reagent (ammoniacal silver nitrate) to:
- (i) detect the presence of an aldehyde group
 - (ii) distinguish between aldehydes and ketones, explained in terms of the oxidation of aldehydes to carboxylic acids with reduction of silver ions to silver.

The equation for this reaction is **not** required. Structure of derivative **not** required.

PAG7 (see also 6.3.1 c)
HSW4 Qualitative analysis.

In equations involving Tollens' reagent, $[\text{O}]$ is acceptable.

PAG7 (see also 6.3.1 c)
HSW4 Qualitative analysis.